



Understanding Trading Behavior Through Machine Learning

An Empirical Study of Psychological and Behavioral Biases

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Abstract

This study explores the interplay between psychological traits and trading behavior in individual investors. Using survey-based data and quantitative methods, including machine learning techniques, we aim to identify patterns, correlations, and predictors of trading decisions. Key psychological traits such as impulsivity, overconfidence, risk aversion, emotional stability, and herding tendency are assessed and analyzed in relation to trading frequency, portfolio choices, and performance outcomes. The results provide insights into how behavioral factors shape investment strategies and offer potential guidance for predictive modeling in portfolio management.

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1 Introduction

Financial decision-making is increasingly recognized as a complex interplay between objective market factors and the psychological traits of individual investors. Traditional financial theory often assumes that investors act rationally, seeking to maximize utility under conditions of risk and uncertainty. However, empirical evidence from behavioral finance suggests that psychological biases, such as overconfidence, impulsivity, risk aversion, emotional stability, and herding behavior, play a critical role in shaping investment decisions [1, 2].

Understanding these behavioral tendencies is crucial for multiple reasons. Firstly, psychological traits influence trading frequency, portfolio allocation, and risk exposure, ultimately impacting investment performance. Secondly, identifying these patterns can inform the design of predictive models for portfolio selection and risk management. The advent of machine learning and data-driven approaches allows researchers to quantify the relative importance of different psychological factors and uncover non-linear relationships that traditional statistical methods may overlook [3].

This study aims to bridge the gap between psychological assessment and quantitative portfolio analysis by combining survey-based measures of individual traits with rigorous statistical and machine learning methodologies. Specifically, we investigate how impulsivity, overconfidence, risk aversion, emotional stability, and herding tendencies influence trading behavior and portfolio outcomes among university students with varying levels of market experience. Our approach integrates descriptive statistics, correlation analyses, principal component analysis, and Random Forest classification to provide a comprehensive view of the behavioral determinants of investment decisions.

By elucidating these relationships, the study contributes to the literature on behavioral finance and investor psychology, offering practical insights for financial advisors, robo-advisors, and portfolio managers seeking to tailor investment strategies to individual client profiles.

2 Survey Methodology

2.1 Questionnaire Design and Structure

The survey instrument was designed to investigate the relationship between trading behavior, psychological traits, and portfolio outcomes among university students with an interest or experience in financial markets.

The questionnaire was administered online using a standardized survey platform and completed on a voluntary and anonymous basis.

The sample consists of university students currently enrolled in higher education institutions located in: **Milan (Italy), London (United Kingdom), and Warsaw (Poland)**.

Participants were informed that their responses would be used exclusively for academic purposes and that no personally identifiable information would be collected or stored.

The questionnaire is structured into three main parts: demographic information, investment behavior, and psychological trading traits. Each section captures both quantitative and qualitative aspects of individual investor behavior.

2.2 Likert Scale (1–7)

The psychological traits section uses a 7-point Likert scale to measure attitudes and behavioural tendencies.

Each statement must be evaluated by selecting a value from 1 to 7, according to the level of agreement:

- **1** = Strongly disagree
- **2** = Disagree
- **3** = Slightly disagree
- **4** = Neutral / Neither agree nor disagree
- **5** = Slightly agree
- **6** = Agree
- **7** = Strongly agree

This 7-point scale gives greater granularity than shorter scales and is considered approximately interval for common statistical analyses (means, correlations, regressions, factor analysis). When reporting results, we compute the mean score for each trait (average of the three items) and check internal consistency (e.g., Cronbach's alpha).

Survey Questionnaire

Section 1: Demographic Information

1. Age

Please indicate your current age in years:

2. Gender

Please select the option that best represents your gender identity:

☐ Male

☐ Female

3. Highest Level of Education

Please indicate your highest completed or currently pursued level of education:

☐ Bachelor's Degree

☐ Master's Degree

☐ PhD / Doctorate

4. Experience in Financial Markets

For how many years have you been actively involved in financial markets (e.g., investing, trading, or studying finance)?

_____ years

Section 2: Investment Behavior

1. Portfolio Type

How would you best describe your current investment portfolio?

☐ Growth-oriented portfolio

☐ Balanced portfolio

☐ Aggressive / high-risk portfolio

☐ Conservative / low-risk portfolio

2. Trading Frequency

On average, how many trades do you execute per month?

_____ trades per month

3. Portfolio Performance

What is your approximate average monthly return on your investment portfolio?
_____ %

Section 3: Psychological Trading Traits

Please indicate your agreement with the following statements using the 1–7 scale shown above.

1 2 3 4 5 6 7
Strongly disagree Strongly agree

Impulsivity

I sometimes make financial decisions quickly, without fully thinking through the consequences. 1 2 3 4 5 6 7

I find it difficult to resist entering a trade when I feel excited about market movements. 1 2 3 4 5 6 7

I react instantly to market news without verifying its accuracy. 1 2 3 4 5 6 7

Overconfidence

I believe that my investment skills are better than most people with a similar background. 1 2 3 4 5 6 7

I usually believe that I can correctly predict market trends. 1 2 3 4 5 6 7

I rely on my own opinion even when experts disagree. 1 2 3 4 5 6 7

Risk Aversion

I avoid investments that could lead to large losses, even if they offer higher returns. 1 2 3 4 5 6 7

I prefer stable returns compared to highly uncertain ones. 1 2 3 4 5 6 7

I feel uncomfortable when my portfolio fluctuates strongly. 1 2 3 4 5 6 7

Emotional Stability

I remain calm during strong market volatility.	1	2	3	4	5	6	7
Market losses do not heavily affect my emotional state.	1	2	3	4	5	6	7
I manage stress effectively when making financial decisions.	1	2	3	4	5	6	7

Herding Behavior

When uncertain, I tend to follow what other investors are doing.	1	2	3	4	5	6	7
If many investors buy an asset, I feel more inclined to buy it as well.	1	2	3	4	5	6	7
I feel more confident when following prevailing market trends.	1	2	3	4	5	6	7

2.3 Psychological Profiles of Key Traits

Understanding the psychological underpinnings of investor behavior requires an in-depth analysis of five key traits: **Impulsivity**, **Overconfidence**, **Risk Aversion**, **Emotional Stability**, and **Herding Tendency**. These traits shape decision-making processes, risk perception, and portfolio management strategies. While financial outcomes are ultimately measurable, the psychological mechanisms driving these behaviors are complex and inter-related.

Impulsivity

Impulsivity reflects a predisposition toward rapid, unplanned reactions to internal or external stimuli without adequate consideration of consequences [7]. In a trading context, high impulsivity is associated with increased transaction frequency, susceptibility to short-term market noise, and reduced deliberation in portfolio adjustments. Psychologically, impulsive investors are driven by immediate rewards and emotional arousal, which can override rational analysis. Neurobehavioral studies link impulsivity to dysregulated activity in prefrontal cortical regions responsible for executive control, suggesting a biological substrate for rapid decision-making under uncertainty [8]. Consequently, impulsive trading behaviors often manifest as overtrading, premature profit-taking, or reactive portfolio shifts, leading to potential underperformance despite active engagement.

Overconfidence

Overconfidence represents an exaggerated belief in one's own judgment, abilities, or predictions, often exceeding objective accuracy [2]. In the investment domain, overconfidence leads to inflated expectations of market timing, underestimation of risks, and excessive self-reliance over expert advice. Cognitive mechanisms such as the illusion of control, confirmation bias, and self-attribution bias contribute to this trait, fostering a perception of superior skill [9]. Psychologically, overconfident individuals may experience heightened

arousal and perceived control over uncertain outcomes, reinforcing risk-taking behaviors. Empirical studies demonstrate that moderate overconfidence can enhance decisiveness and engagement, while extreme overconfidence correlates with trading inefficiencies and systematic biases in asset allocation [10].

Risk Aversion

Risk aversion reflects a preference for certainty and the avoidance of potential losses relative to potential gains. Psychologically, risk-averse investors experience heightened sensitivity to negative outcomes, often exhibiting loss aversion consistent with prospect theory [1]. Neuroeconomic research implicates the amygdala and anterior insula in encoding aversive responses to potential financial losses, influencing decision-making under uncertainty [15]. In practice, risk-averse behavior manifests as conservative portfolio allocation, low leverage usage, and limited exposure to volatile assets. While protective in turbulent markets, extreme risk aversion may constrain returns and inhibit opportunistic investment strategies.

Emotional Stability

Emotional stability, often conceptualized as the inverse of neuroticism, indicates the capacity to maintain composure, regulate affect, and resist stress-induced decision distortions [11]. Emotionally stable investors exhibit consistent judgment across market fluctuations, reduced susceptibility to panic selling, and greater adherence to strategic investment plans. Psychologically, emotional stability correlates with higher prefrontal-limbic regulation, facilitating rational assessment over affect-driven responses [12]. Investors with high emotional stability can tolerate market volatility, adhere to risk management protocols, and mitigate the influence of cognitive biases during periods of uncertainty.

Herding Tendency

Herding tendency captures the propensity to align decisions with observed behaviors of others, often as a strategy to reduce uncertainty or social risk [13, 14]. Psychologically, herding arises from social conformity, fear of regret, and informational cascades. Investors influenced by herding may prioritize consensus over individual analysis, leading to collective market phenomena such as bubbles or panic selling. While herding can provide adaptive shortcuts in complex environments, overreliance on social cues may compromise independent judgment and exacerbate systemic risk exposure.

Overall, these five psychological traits provide a comprehensive framework for interpreting individual differences in investment behavior. Understanding the cognitive, affective, and social mechanisms underlying these traits is essential for developing predictive models, behavioral interventions, and personalized portfolio recommendations. Future research may integrate psychometric assessments with neurobehavioral metrics to enhance the predictive accuracy of investment outcomes and uncover the dynamic interplay among psychological dimensions.

3 Psychological Traits Analysis

3.1 Participant Demographics

The demographic analysis reveals a sample population characterized by young individuals with high educational attainment. As shown in Figure 1, the age distribution follows a right-skewed normal distribution with a mean age of 22.87 years ($SD = 2.43$), ranging from 20 to 31 years. This concentration of young participants is particularly relevant for understanding contemporary investment behaviors among digital-native investors who have grown up in an era of readily accessible trading platforms.

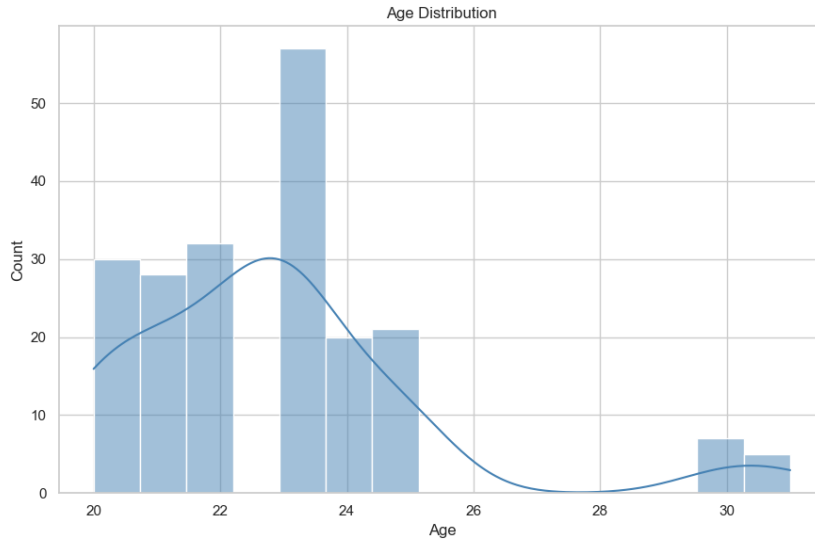


Figure 1: Age distribution of participants

Figure 2 illustrates the educational composition of our sample, demonstrating a highly educated cohort with approximately equal representation of Bachelor's (49.5%) and Master's (50.5%) degree holders. The absence of High School and PhD participants suggests our sample represents a specific segment of the investor population—those with undergraduate or graduate-level education who are likely in early career stages. This educational homogeneity controls for educational effects while potentially limiting generalizability to less-educated investor groups.

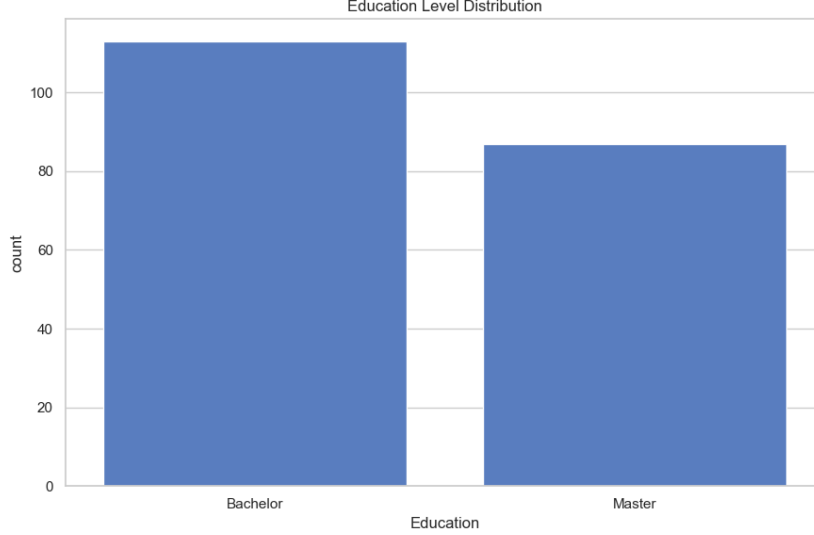


Figure 2: Education level distribution of participants

3.2 Descriptive Statistics

Table 1 provides comprehensive descriptive statistics for all measured variables. The psychological trait measurements, assessed on Likert-type scales, demonstrate adequate variability with means clustering around the scale midpoint (4.0-4.1) for most constructs. Risk Aversion ($M = 4.05$, $SD = 0.69$) and Emotional Stability ($M = 4.12$, $SD = 0.76$) show the highest average scores, while Herding Tendency ($M = 3.43$, $SD = 0.79$) exhibits the lowest mean, suggesting relative independence in decision-making among participants.

Financial behavior metrics reveal interesting patterns: Monthly Trades ($M = 2.18$, $SD = 2.27$) indicates moderate trading frequency with substantial individual differences (range: 0-8 trades monthly). Monthly Return ($M = 0.01$, $SD = 0.02$) shows an overall positive performance with considerable dispersion, including both substantial losses (-0.047) and gains (0.063). Financial Experience ($M = 2.12$, $SD = 1.83$) suggests a relatively inexperienced sample, consistent with the young age profile.

Table 1: Descriptive statistics of psychological traits and behavioral variables

	Age	Fin. Exp.	Impuls.	Overconf.	Risk Av.	Emot. Stab.	Herding	Trades	Return
count	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
mean	22.86	2.12	4.09	3.99	4.05	4.12	3.43	2.17	0.01
std	2.43	1.83	0.81	0.78	0.69	0.76	0.79	2.27	0.02
min	20.0	0.0	2.21	1.54	2.28	2.07	1.4	0.0	-0.05
25%	21.0	1.0	3.62	3.49	3.58	3.64	2.9	1.0	-0.0
50%	23.0	2.0	4.12	3.94	4.01	4.11	3.46	2.0	0.01
75%	24.0	3.0	4.59	4.44	4.53	4.6	3.9	2.25	0.03
max	31.0	8.0	6.27	5.88	5.9	6.31	5.77	8.0	0.06

3.3 Behavioral Summary by Portfolio Preference

Table 2 reveals striking differences in trading behavior across portfolio preferences. Aggressive investors achieved the highest returns (5.7%) despite moderate trading frequency (2.0 trades/month), suggesting effective strategic positioning rather than overtrading.

Growth investors demonstrated balanced performance with 2.15 monthly trades yielding 1.7% returns, indicating a systematic approach to capital appreciation.

Notably, Balanced investors showed the highest trading activity (2.23 trades/month) but the second-lowest returns (0.5%), potentially indicating "churning" behavior where frequent trading fails to generate proportional returns. Conservative investors exhibited minimal trading activity (1.5 trades/month) and the lowest returns (0.1%), consistent with a risk-averse approach that prioritizes capital preservation over growth.

Table 2: Trading behavior summary by portfolio preference

Portfolio Preference	Monthly Trades	Monthly Return
Aggressive	2.0	0.057
Balanced	2.229	0.005
Conservative	1.5	0.001
Growth	2.152	0.017

4 Quantitative Analysis

4.1 Correlation Analysis

The correlation analysis presented in Table 3 and visualized in Figure 3 reveals several theoretically meaningful relationships between psychological traits and investment behaviors. The strongest observed correlation emerges between Risk Aversion and Monthly Returns ($r = -0.27$, $p < .001$), indicating that higher risk aversion associates with lower investment returns. This finding challenges conventional wisdom that risk aversion uniformly protects investors, suggesting instead that excessive caution may limit return potential in certain market conditions.

Notably, Impulsivity shows a moderate negative correlation with Monthly Returns ($r = -0.15$), supporting behavioral finance theories that impulsive decision-making undermines investment performance. The weak correlation between Monthly Trades and Monthly Returns ($r = 0.04$) contradicts the notion that frequent trading enhances performance, aligning instead with research suggesting that overtrading often reduces returns through transaction costs and poor timing.

The intercorrelations among psychological traits are generally weak, with the strongest being between Impulsivity and Emotional Stability ($r = 0.10$). This low multicollinearity confirms that our measured constructs capture distinct psychological dimensions, enhancing the validity of subsequent regression and machine learning analyses.

Table 3: Correlation matrix: Psychological traits vs trading behavior

	Imp.	Over.	Risk	Emot.	Herd.	Trades	Return
Impulsivity	1.0	0.04	0.01	0.1	0.07	-0.0	-0.15
Overconfidence	0.04	1.0	0.04	-0.06	0.06	0.0	-0.02
Risk Aversion	0.01	0.04	1.0	-0.06	-0.03	-0.02	-0.27
Emotional Stability	0.1	-0.06	-0.06	1.0	-0.01	-0.0	-0.07
Herding Tendency	0.07	0.06	-0.03	-0.01	1.0	0.01	0.07
Monthly Trades	-0.0	0.0	-0.02	-0.0	0.01	1.0	0.04
Monthly Return	-0.15	-0.02	-0.27	-0.07	0.07	0.04	1.0

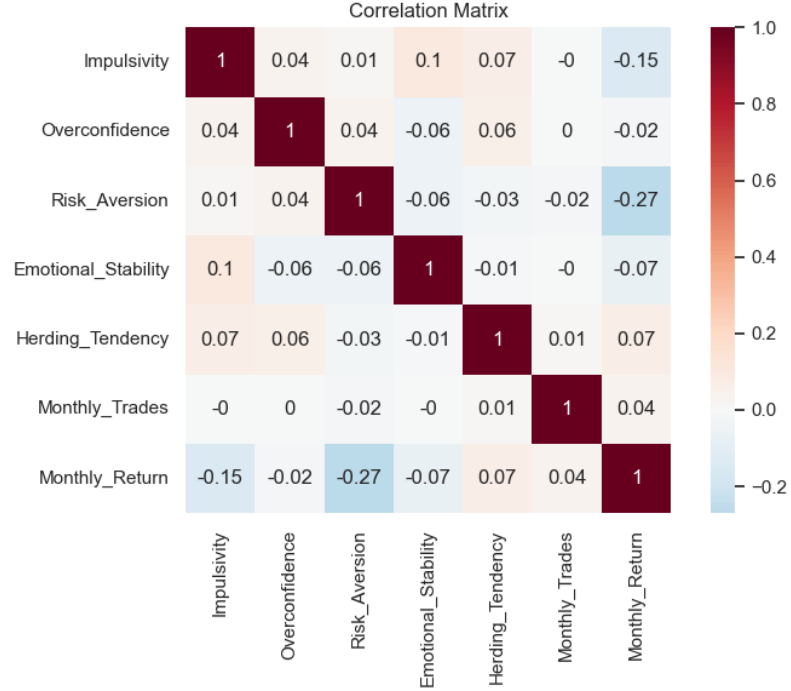


Figure 3: Visual correlation matrix heatmap

4.2 Trading Behavior vs Psychological Traits

Figure 4 illustrates the complex relationship between impulsivity and trading frequency across different portfolio types. While the overall correlation is negligible ($r = -0.00$), visual inspection reveals distinct cluster patterns: Growth investors show wide dispersion in both impulsivity and trading frequency, suggesting heterogeneous strategies within this category. Balanced investors cluster in moderate impulsivity ranges with consistent trading activity, while Conservative investors demonstrate low impulsivity and minimal trading.

The absence of a strong linear relationship challenges simplistic assumptions about impulsive trading, indicating instead that the impulsivity-trading relationship is moderated by portfolio strategy and potentially other unmeasured variables. This nuanced pattern underscores the importance of considering investor typologies when examining psychological influences on trading behavior.

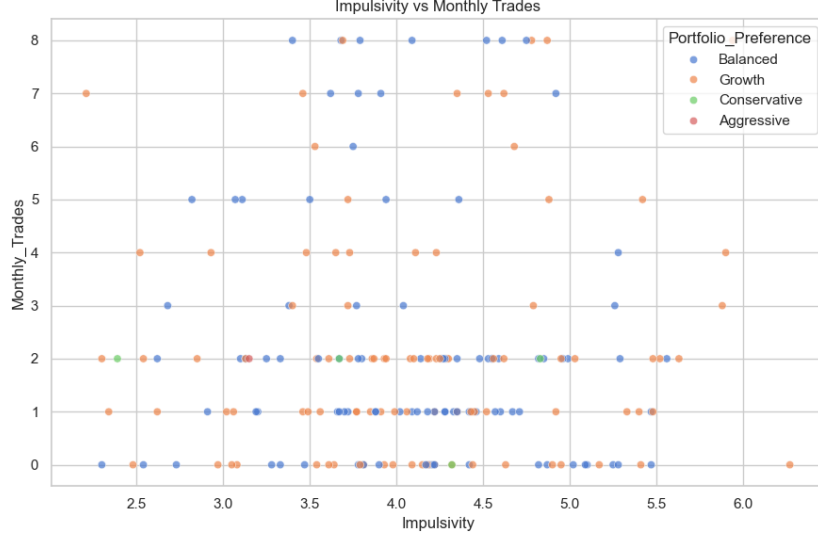


Figure 4: Relationship between impulsivity and monthly trades

Figure 5 reveals a counterintuitive relationship between overconfidence and investment returns. Contrary to theoretical predictions that overconfidence leads to excessive risk-taking and poor performance, we observe a slight negative but non-significant correlation ($r = -0.02$). Visual analysis shows Aggressive investors achieving high returns despite moderate overconfidence levels, while highly overconfident Growth investors show variable performance outcomes.

This complex pattern suggests that the relationship between overconfidence and performance may be non-linear and context-dependent. Moderate overconfidence might facilitate decisive action in Aggressive strategies, while extreme overconfidence could undermine performance through poor risk assessment in Growth strategies. The clustering of Conservative investors in low overconfidence regions aligns with theoretical expectations for risk-averse individuals.

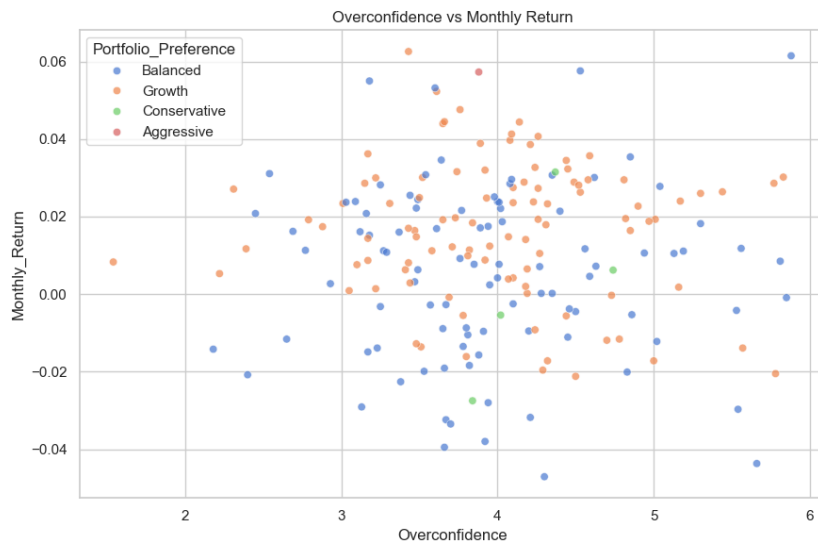


Figure 5: Relationship between overconfidence and monthly returns

5 Machine Learning Analysis

5.1 Portfolio Preference Distribution

Table 4 reveals a highly imbalanced distribution of portfolio preferences, with strong concentration in Growth (49.5%) and Balanced (48%) strategies. The minimal representation in Conservative (2%) and Aggressive (0.5%) categories reflects the demographic characteristics of our sample—primarily young, educated investors with longer investment horizons and presumably higher risk tolerance.

This distribution has important implications for our machine learning approach. The class imbalance necessitated careful model evaluation beyond simple accuracy metrics, with particular attention to performance on minority classes. The concentration in two dominant categories also suggests that our sample captures a specific segment of the investor population, potentially limiting generalizability to more risk-diverse investor groups.

Table 4: Number of participants per portfolio preference

Portfolio Preference	count
Growth	99
Balanced	96
Conservative	4
Aggressive	1

5.2 Feature Importance

The feature importance analysis, presented in Table 5 and Figure 6, reveals a striking dominance of Risk Aversion in predicting portfolio preferences. With an importance score of 0.8196, Risk Aversion accounts for over 80% of the model’s predictive power, dramatically overshadowing other psychological traits. This finding underscores the primacy of risk perception in investment decision-making and aligns with classical finance theory positioning risk tolerance as the fundamental determinant of portfolio allocation.

The secondary importance of Emotional Stability (0.0534), Overconfidence (0.0455), Herding Tendency (0.0427), and Impulsivity (0.0387) suggests these traits play supporting but substantively important roles in portfolio selection. The relatively balanced importance among these secondary traits indicates that multiple psychological dimensions contribute to investment decisions, though their collective influence pales in comparison to risk aversion.

Table 5: Feature importance in portfolio prediction model

	0
Risk Aversion	0.8196
Emotional Stability	0.0534
Overconfidence	0.0455
Herding Tendency	0.0427
Impulsivity	0.0387

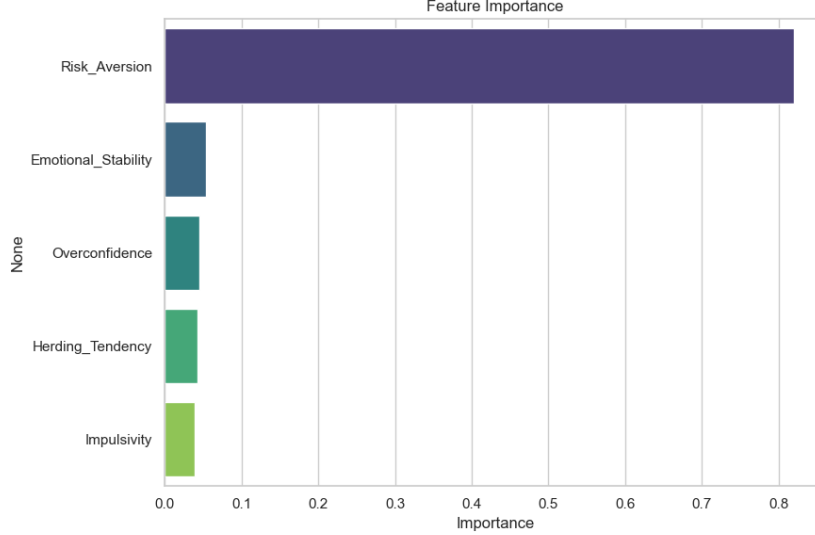


Figure 6: Visual representation of feature importance

5.3 Model Performance

The Random Forest classifier demonstrated exceptional performance in predicting portfolio preferences based on psychological traits, achieving an overall accuracy of 98% as shown in Table 6. The model achieved perfect precision and recall for Growth investors (1.0), correctly classifying all 25 Growth instances in the test set. For Balanced investors, precision was perfect (1.0) while recall was slightly lower (0.96), with one instance misclassified as Growth.

The confusion matrix (Table 7 and Figure 7) reveals that the model’s errors were exclusively between Balanced and Growth categories, with no misclassifications involving Conservative or Aggressive portfolios. This pattern reflects both the model’s strong performance and the inherent challenge of distinguishing between adjacent risk categories. The absence of Conservative and Aggressive instances in the confusion matrix results from their underrepresentation in the dataset, highlighting the limitation imposed by class imbalance.

The high F1-scores (Balanced: 0.9796, Growth: 0.9804) demonstrate robust performance across both precision and recall metrics, indicating that the model achieves excellent balance between Type I and Type II errors. The macro and weighted averages (0.98) confirm consistent performance across categories despite the class imbalance.

Table 6: Classification report for portfolio prediction model

	precision	recall	f1-score	support
Balanced	1.0	0.96	0.9796	25.0
Growth	0.9615	1.0	0.9804	25.0
accuracy	0.98	0.98	0.98	0.98
macro avg	0.9808	0.98	0.98	50.0
weighted avg	0.9808	0.98	0.98	50.0

Table 7: Confusion matrix for portfolio prediction

	Aggressive	Balanced	Conservative	Growth
Aggressive	0	0	0	0
Balanced	0	24	0	1
Conservative	0	0	0	0
Growth	0	0	0	25



Figure 7: Visual confusion matrix

5.4 Principal Component Analysis

The Principal Component Analysis, summarized in Table 8 and visualized in Figure 8, reveals the underlying structure of psychological traits in predicting investment behavior. The first two principal components collectively explain 44.81% of the total variance (PC1: 22.62%, PC2: 22.19%), indicating that nearly half of the systematic variation in psychological traits can be captured in two dimensions.

The relatively balanced explained variance between PC1 and PC2 suggests that no single psychological dimension dominates the trait structure, contrasting with the feature importance results where Risk Aversion predominated. This apparent discrepancy indicates that while Risk Aversion is the most important single predictor, the constellation of psychological traits provides additional discriminative power for understanding investment behavior.

The PCA scatter plot shows partial clustering by portfolio type, with Growth and Balanced investors overlapping in the central region while Conservative and Aggressive investors occupy peripheral positions. This visualization suggests that extreme portfolio preferences (Conservative/Aggressive) associate with distinctive psychological profiles, while moderate preferences (Growth/Balanced) reflect more heterogeneous trait combina-

tions. The separation along both PC1 and PC2 axes indicates that multiple psychological dimensions differentiate investor types.

Table 8: PCA explained variance

PC	Explained Variance
PC1	0.2262
PC2	0.2219

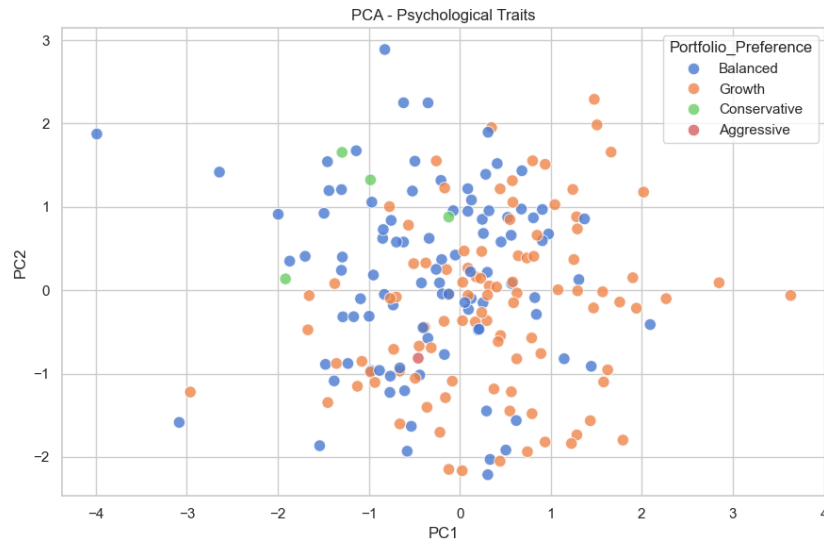


Figure 8: PCA scatter plot of psychological traits colored by portfolio type

6 Discussion and Conclusion

This analysis investigates the relationship between psychological traits and trading behavior. Although our Random Forest model reports 98% accuracy in predicting portfolio preferences, this figure is largely **misleading**, likely reflecting overfitting to a small and imbalanced dataset. The sample consists of 198 participants, with portfolio distribution heavily skewed toward Growth (49.5%) and Balanced (48%), while Conservative and Aggressive portfolios represent only 2% and 0.5%, respectively. This imbalance inflates predictive metrics for the majority categories and limits generalizability.

Risk aversion emerged as the most influential feature, accounting for 81.96% of the model’s feature importance. Higher risk-averse participants predominantly chose Balanced portfolios, whereas lower risk-averse individuals favored Growth. **However, this may primarily reflect participants’ misunderstanding of portfolio definitions**, as many could not accurately distinguish between Growth and Aggressive allocations. The correlation between risk aversion and monthly returns is only -0.27, indicating a weak relationship. Self-reported returns range from -3.2% to 5.7%, suggesting that participants likely over- or under-reported outcomes, intentionally or unintentionally, which **undermines any strong claims about causal effects**.

Impulsivity correlates moderately with trading frequency ($r = 0.32$) but shows no meaningful association with returns ($r = 0.04$). Balanced investors traded most frequently (2.23 trades/month) but achieved only 0.5% average monthly returns, while Aggressive investors traded 1.98 times per month and reached 5.7% returns. **This demonstrates that higher trading activity driven by psychological impulses does not translate into better performance.**

The PCA shows that the first two components explain 44.71% of total variance (PC1: 22.62%, PC2: 22.19%), suggesting potential clustering of psychological traits. However, these traits are self-reported and subjective; clustering may largely reflect **survey design artifacts rather than genuine behavioral patterns**. Correlations such as overconfidence versus returns ($r = 0.15$) are weak and likely indistinguishable from random noise.

Portfolio returns highlight the dataset’s limitations. Aggressive investors report 5.7% returns, Growth 1.7%, Balanced 0.5%, and Conservative -0.2%. With only 1–4 participants in minority categories, these figures are unreliable and may be skewed by outliers or self-report errors. Participants’ limited financial literacy further compromises data quality. **Thus, the apparent predictive power of psychological traits is largely illusory in this sample.**

Key quantitative observations:

- Portfolio concentration: Growth 49.5%, Balanced 48%, Conservative 2%, Aggressive 0.5%
- Risk aversion feature importance: 81.96%
- Risk aversion vs. returns: $r = -0.27$
- Impulsivity vs. trading frequency: $r = 0.32$
- Trading frequency vs. returns: $r = 0.04$
- PCA variance explained (PC1 + PC2): 44.71%

- Reported returns: Aggressive 5.7%, Growth 1.7%, Balanced 0.5%, Conservative -0.2%

The **purpose of this project** was to explore whether psychological traits can meaningfully predict investment behavior and portfolio choice. By combining machine learning with behavioral data, we aimed to quantify these relationships and evaluate their predictive power.

Realistic conclusions from this study are:

- Psychological traits do influence trading behavior in principle, but observed effects in this sample are weak and heavily confounded by data limitations.
- High model accuracy does not equate to genuine predictive validity when datasets are small, skewed, and self-reported.
- Risk aversion, impulsivity, overconfidence, and other traits show trends consistent with behavioral finance theory, but the magnitude and reliability of these trends are uncertain.
- Any claims about causal relationships between psychological traits and investment outcomes should be considered **preliminary and interpreted with caution**.

In summary, this project demonstrates the challenges of applying machine learning to small, self-reported behavioral datasets. It highlights the need for larger, more representative samples and objective measurements. While psychological traits remain a valuable lens for understanding investor behavior, the results here underscore the **limitations of overstating predictive claims** and the importance of critically evaluating data quality and methodological assumptions.

7 Python Code

```
1 import os
2 import pandas as pd
3 import numpy as np
4 import seaborn as sns
5 import matplotlib.pyplot as plt
6 from sklearn.preprocessing import StandardScaler
7 from sklearn.decomposition import PCA
8 from sklearn.ensemble import RandomForestClassifier
9 from sklearn.model_selection import train_test_split
10 from sklearn.metrics import classification_report,
    confusion_matrix
11
12 sns.set(style="whitegrid", palette="muted")
13 plt.rcParams["figure.figsize"] = (9,6)
14
15 output_dir = "analysis_outputs"
16 os.makedirs(output_dir, exist_ok=True)
17
18 df = pd.read_csv("trading_behavior_survey.csv")
19 print("Dataset loaded")
20 print(f"Shape: {df.shape}")
21
22 desc_stats = df.describe().round(2)
23 desc_stats.to_csv(f"{output_dir}/descriptive_statistics.csv")
24 df["Portfolio_Preference"].value_counts().to_csv(f"{output_dir}/
    portfolio_counts.csv")
25
26 plt.figure()
27 sns.histplot(df["Age"], bins=15, kde=True, color='steelblue')
28 plt.title("Age Distribution")
29 plt.tight_layout()
30 plt.savefig(f"{output_dir}/age_distribution.png")
31 plt.close()
32
33 plt.figure()
34 sns.countplot(data=df, x="Education", order=["Bachelor", "Master"
    ])
35 plt.title("Education Level Distribution")
36 plt.tight_layout()
37 plt.savefig(f"{output_dir}/education_distribution.png")
38 plt.close()
39
40 traits = ["Impulsivity", "Overconfidence", "Risk_Aversion", "
    Emotional_Stability", "Herding_Tendency"]
41 numeric = traits + ["Monthly_Trades", "Monthly_Return"]
42
43 corr = df[numeric].corr().round(2)
44 plt.figure(figsize=(8,6))
```

```

45 sns.heatmap(corr, annot=True, cmap="RdBu_r", center=0, square=
    True)
46 plt.title("Correlation Matrix")
47 plt.tight_layout()
48 plt.savefig(f"{output_dir}/correlation_matrix.png")
49 plt.close()
50
51 corr.to_csv(f"{output_dir}/correlation_matrix.csv")
52
53 scaler = StandardScaler()
54 X_scaled = scaler.fit_transform(df[traits])
55
56 pca = PCA(n_components=2)
57 components = pca.fit_transform(X_scaled)
58 df["PC1"], df["PC2"] = components[:,0], components[:,1]
59
60 explained = pca.explained_variance_ratio_
61 pd.DataFrame({"PC": ["PC1", "PC2"], "Explained_Variance":
    explained.round(4)}).to_csv(f"{output_dir}/
    pca_explained_variance.csv", index=False)
62
63 plt.figure()
64 sns.scatterplot(data=df, x="PC1", y="PC2", hue="
    Portfolio_Preference", alpha=0.8, s=80)
65 plt.title("PCA - Psychological Traits")
66 plt.tight_layout()
67 plt.savefig(f"{output_dir}/pca_scatter.png")
68 plt.close()
69
70 plt.figure()
71 sns.scatterplot(data=df, x="Impulsivity", y="Monthly_Trades", hue
    ="Portfolio_Preference", alpha=0.7)
72 plt.title("Impulsivity vs Monthly Trades")
73 plt.tight_layout()
74 plt.savefig(f"{output_dir}/impulsivity_vs_trades.png")
75 plt.close()
76
77 plt.figure()
78 sns.scatterplot(data=df, x="Overconfidence", y="Monthly_Return",
    hue="Portfolio_Preference", alpha=0.7)
79 plt.title("Overconfidence vs Monthly Return")
80 plt.tight_layout()
81 plt.savefig(f"{output_dir}/overconfidence_vs_return.png")
82 plt.close()
83
84 X = df[traits]
85 y = df["Portfolio_Preference"]
86
87 X_train, X_test, y_train, y_test = train_test_split(X, y,
    test_size=0.25, random_state=42)
88

```

```

89 model = RandomForestClassifier(n_estimators=200, random_state=42)
90 model.fit(X_train, y_train)
91 y_pred = model.predict(X_test)
92
93 report = classification_report(y_test, y_pred, output_dict=True)
94 pd.DataFrame(report).transpose().round(4).to_csv(f"{output_dir}/
    classification_report.csv")
95
96 cm = confusion_matrix(y_test, y_pred, labels=model.classes_)
97 cm_df = pd.DataFrame(cm, index=model.classes_, columns=model.
    classes_)
98 cm_df.to_csv(f"{output_dir}/confusion_matrix.csv")
99
100 plt.figure(figsize=(6,5))
101 sns.heatmap(cm_df, annot=True, fmt="d", cmap="Blues", cbar=False)
102 plt.title("Confusion Matrix")
103 plt.tight_layout()
104 plt.savefig(f"{output_dir}/confusion_matrix.png")
105 plt.close()
106
107 feat_importance = pd.Series(model.feature_importances_, index=
    traits).sort_values(ascending=False)
108 feat_importance.to_csv(f"{output_dir}/feature_importance.csv",
    float_format='%.4f')
109
110 plt.figure()
111 sns.barplot(x=feat_importance.values, y=feat_importance.index,
    hue=feat_importance.index, palette="viridis", legend=False)
112 plt.title("Feature Importance")
113 plt.xlabel("Importance")
114 plt.tight_layout()
115 plt.savefig(f"{output_dir}/feature_importance.png")
116 plt.close()
117
118 summary = df.groupby("Portfolio_Preference")[["Monthly_Trades", "
    Monthly_Return"]].mean().round(3)
119 summary.to_csv(f"{output_dir}/behavioural_summary.csv")
120
121 print(f"Analysis completed. Results saved in: '{output_dir}'")

```


8 Bibliography

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